

Software Engineering and Data Science
SEIS 766: Vision AI
Assignment #4

Due Date: March 12th

Write a MatLab or Python program to identify “yolk” in the zebrafish embryo images. The images are zipped in “**FishEmbryo_All_Images.zip**” on Canvas example folder: “**File > Course Examples + data > VAI_Slide_SemanticSegmentation**”. For this assignment, use the “Yolk” subfolder of “FishEmbryo_All_Images.zip\FishEmbryo_Raw_Images_Yolk” and “FishEmbryo_All_Images.zip\FishEmbryo_Label_Images_...”.

1. For this assignment, you can use any kind of CNN object identification model, including your own CNN or any pretrained model.
2. You can either split images into training + testing, or work on this assignment by using all the images as training images without testing images.
3. Pick 3 raw embryo images and display each raw image with its corresponding “yolk” detection result. You can display each pair of images in two separate figures, or you can overlap each pair together and display the overlapped image in one figure.
4. What is the IOU (Intersection Over Union) of your detection result from all your either training or testing images.

Submission Guideline:

1. Please include the answers to the above questions in a WORD document. Please also put **your name** on the top of your WORD document.
2. Please print your program (matlab or python) as **PDF / html** and include the **PDF / html** in your submission. Please name your program as “a4.m/.mlx/.py/.inpyb”.
3. Please also include your program in the formats like .m/.mlx/.py/.inpyb in your submission.
4. Prepare EVERYTHING mentioned in the guideline and submit them on **Canvas** no later than the due date. Please do **NOT** zip your files.
5. Please carefully follow the submission guideline. Otherwise, the instructor may not be able to grade your assignment.